

Candidate Name: _____

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Preliminary Examination II 2009
Pre-University 3

MATHEMATICS 9740
PAPER 1

9740/01

Thursday

17 September 2009

3 hours

Additional materials:
Cover Page
Writing paper
MF15

READ THESE INSTRUCTIONS FIRST

Write your name, admission number and class in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This question paper consists of **5** printed pages.

[Turn over]

- 1 Solve the inequality

$$\frac{x+4}{x^2-4} \geq x. \quad [4]$$

- 2 (i) Express $\frac{2}{n(n-1)}$ in partial fractions. [2]

- (ii) Hence, find $\sum_{r=2}^N \frac{1}{r(r-1)}$ in terms of N , and deduce the value of $\sum_{r=2}^{\infty} \frac{1}{r(r-1)}$. [3]

- 3 Due to the poor economy, a chocolate factory would like to cut the production of one type of chocolate bars. The cost and profit per bar are shown in the following table.

	Cost per bar (\$)	Profit per bar (\$)
Dark Chocolate	5.20	4.80
Milk Chocolate	3.10	2.90
White Chocolate	2.50	1.50

In the month of August 2009, total cost for producing 800 chocolate bars is \$2540. If the company makes a total profit of \$2160 and all the chocolate bars that were produced were successfully sold, determine the number of chocolate bars sold per type by using the product of two matrices.

Which type of chocolate bar should the factory stop producing?
Give a reason for your choice.

[5]

- 4 An arithmetic series has first term a and common difference d , and a geometric series has common ratio r , where a and d are non-zero. The first three terms of the geometric series are equal to the first, fifth and seventh terms respectively of the arithmetic series.

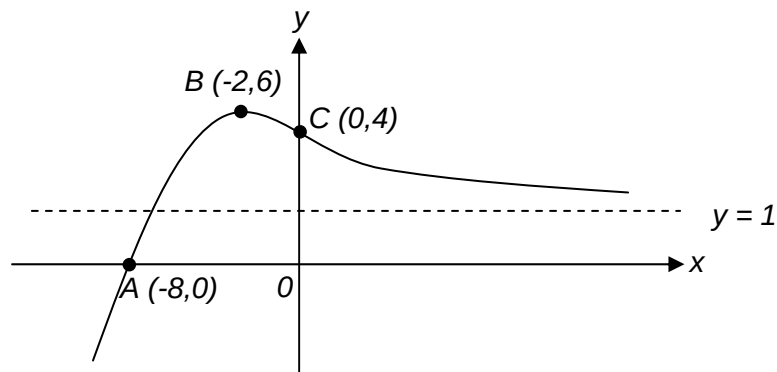
- (i) Show that $2r^2 - 3r + 1 = 0$. [4]

- (ii) Deduce that the geometric series is convergent. [2]

- (iii) The sum of the first n terms of the geometric series is denoted by S_n . Find S_n in terms of a and n . [2]

- 5 (a) The curve C has equation $y = \frac{3x+2}{x-4}$.
- (i) Write down the equations of all the asymptotes of C . [2]
- (ii) Sketch the curve C , indicating all turning points, asymptotes and axial intercepts, if any. [3]
- (iii) Find the set of values of k for which the line $y = kx$ intersects C at 2 points. Leave your answers to 3 significant figures. [4]

(b)



The diagram shows a sketch of the graph of $y = f(x)$.

On separate diagrams, sketch the graph of

- (i) $y = f(2x+1)$ and [3]
- (ii) $y = f'(x)$. [3]

- 6 (a) Find an expression for $\frac{\sin 2x}{\sqrt{\cos x}}$ given that x is small enough such that x^6 and higher powers of x can be neglected. [4]
- (b) An inverted cone with radius 3 cm and height 6 cm is filled to the brim with water. When the drop in water level is h cm, show that the volume of water in the cone is given by $V = \frac{1}{3}\pi(3 - \frac{1}{2}h)^2(6 - h)$.

Water is leaking from the tip of the cone at a rate of $1\text{cm}^3\text{s}^{-1}$. Find the exact rate of drop in water level of the cone when $h = 1$. [5]

- 7 The line l passes through the points $A(1, 2, 3)$ and $B(1, 4, 7)$. The plane π_1 has equation $3x - 4y + z = 11$. Find
- (i) the acute angle between the line l and the plane π_1 , [5]
 - (ii) the foot of perpendicular from point A to the plane π_1 . [4]
 - (iii) the line of intersection between the plane π_1 and the plane π_2 defined by $x + y + z = 1$. [3]
- 8 A curve has parametric equations $x = t^3, y = \ln t$, for $1 \leq t \leq 2$.
- (i) Show that $\frac{dy}{dx} = \frac{1}{3x}$. [2]
 - (ii) Sketch the curve. [2]
 - (iii) Find the equation of the tangent to the curve at the point where $t = \frac{3}{2}$.
The tangent to the curve at P meets the x -axis and the y -axis at A and B respectively. The origin is denoted by O . Find the area of triangle AOB . [5]
- 9 The sequence $u_1, u_2, u_3, \dots$ is such that $u_1 = 1$ and $u_{n+1} = \frac{u_n}{u_n + 1}$, for all $n \geq 1$.
- (i) Write down the values of u_2, u_3 and u_4 . [3]
 - (ii) Hence, make a conjecture for u_n in terms of n . [1]
 - (iii) Use the method of mathematical induction to prove that your conjecture is true for all positive integers n . [5]

- 10** (a) (i) Solve the equation $z^4 = -16i$, expressing the solutions in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [4]
- (ii) Show all the solutions on an Argand diagram. [2]
- (b) Given that $1 + 2i$ is a root of the equation

$$z^3 + az^2 + bz + 2 = 0,$$

 find the values of the real numbers a and b . [5]
- 11** (a) (i) Without the use of a graphic calculator, find the exact value of $\int_0^{\frac{\pi}{2}} \sin^2 x \, dx$. [3]
- (ii) Find $\int e^x \sin x \, dx$. [4]
- (b) (i) Find the general solution of the differential equation

$$\left(\frac{x^2 + 1}{x} \right) \frac{dy}{dx} = 2.$$
 [2]
- (ii) Find the particular solution of the differential equation in (i) for which $y = 0$ and $x = 0$. [1]
- (iii) Sketch, on a single diagram, three graphs from the same family of curves, including the particular solution found in part (ii). [2]
- (iv) Describe the gradient of every solution curve as $x \rightarrow +\infty$. [1]

END OF PAPER